Style Transfer for Free: Native Same-State Score Residuals in Frozen Rectified-Flow Transformers

Anonymous submission

Abstract

Training-free style transfer on modern text-to-image diffusion models typically bolts on attention-feature replacement, learned adapters, or hand-designed transport heuristics. We show that on a frozen rectified-flow transformer (FLUX.2 [klein] 9B) a single, minimal signal suffices: the *native same-state image-CFG score residual* – the difference between the model's score with and without a style reference as an image condition, evaluated at the *same* noise state – used to optimize only the current latent at inference (AdamW, a few iterations per denoising step). We train nothing and add no modules. Across a 40-style benchmark (91 matched cells, multi-seed) the method produces sharp, content-preserving transfers over radically different styles (impasto, pop-art, engraving, woodcut, anime, expressionism). Crucially, we ablate three "obvious" additions – a content-affinity transport carrier, timestep-scheduled injection strength, and the latent-optimizer choice – and find that none improves over the plain score residual beyond seed noise: the minimal form is deliberate, not incidental. We further show that the standard CSD style metric systematically *under-rates* visually excellent transfers when the reference's subject differs from the target content, corroborating a recent calibration critique, and we report a calibration-aware CSD+ readout alongside qualitative evidence.

Introduction

Diffusion style transfer without training is attractive but crowded with machinery: attention-feature replacement, IP-Adapter-style learned encoders, and transport or AdaIN heuristics. Each adds assumptions, parameters, or reference-coupling to the base model. We ask how little is actually needed on a modern *rectified-flow* transformer, whose straight-line ODE and native image-conditioning path differ from the DDPM U-Nets most prior training-free methods target.

Our answer is a single native signal. A frozen model that supports reference-image conditioning already "knows" how the style reference should bend its prediction: the score difference between conditioning on the reference image and not, taken at the *same* noise state, is a style-transport direction. We push the current latent along this direction, optimizing only that latent – no weights, no modules – for a few AdamW steps per denoising step. That is the whole method.

This minimalism is not a default we settle for; it is a finding. We tried the natural next step – a content-affinity carrier that reweights the residual toward the current image's own tokens, in the spirit of transport-based style methods – and, across five independent probes (a matched ablation matrix, a carrier-strength sweep, a timestep-schedule sweep, a calibrated-metric check, and a perceptual-change diagnostic), it never produces a robust quality gain. Timestep scheduling of the injection strength and the specific latent-optimizer choice are likewise inert beyond ordinary seed variance. Reporting these as negative results, rather than quietly shipping only the version that "worked," is itself part of the paper's contribution: it tells a practitioner exactly which knobs are worth building and which are not.

Contributions. (i) A minimal training-free style-transfer method for frozen rectified-flow transformers using only the native same-state image-CFG score residual, optimizing the latent alone. (ii) A rigorous negative-ablation study showing that a content-affinity carrier, timestep scheduling, and the latent-optimizer choice each add nothing beyond seed noise – the minimal form is sufficient. (iii) A 40-style multi-seed benchmark and a calibration analysis demonstrating that raw CSD under-rates visually excellent transfers, with a CSD+ readout and qualitative evidence.

Related Work

Training-free style transfer in diffusion. Classical neural style transfer matches feature statistics or adaptive instance-normalization moments (Gatys, Ecker, and Bethge 2016; Huang and Belongie 2017). In diffusion models, training-free style transfer typically replaces or shares self-attention features between a reference and a generation (Hertz et al. 2024; Chung, Hyun, and Heo 2024), building on attention-control machinery originally designed for text-driven editing (Hertz et al. 2023; Tumanyan et al. 2023; Yang et al. 2023). A second family learns a dedicated reference-conditioning interface: an image adapter (Ye et al. 2023), a block-selective injection scheme (Wang et al. 2024), or a fine-tuned style token (Sohn et al. 2023). A third family treats style transfer as an optimization or transport problem, using a stochastic-optimal-control formulation over a reference descriptor (Rout

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et al. 2025b) or a score-difference objective in the lineage of score distillation sampling (Poole et al. 2023; Hertz, Aberman, and Cohen-Or 2023; Nam et al. 2024). Each family adds machinery – replaced attention features, a trained encoder, or a hand-designed transport rule. We add none of it: the transport signal is the frozen model's own image-conditioned score contrast, and the only thing we optimize is the current latent.

Test-time guidance. Test-time guidance methods reshape a diffusion sampling trajectory without retraining. Universal and energy-based guidance steer sampling with an auxiliary loss gradient (Bansal et al. 2023); perturbed- and smoothed-attention guidance degrade an internal self-attention map to synthesize an implicit low-quality branch for contrastive guidance (Ahn et al. 2024; Hong 2024); condition-annealed sampling anneals the conditioning signal itself to trade fidelity for diversity (Sadat et al. 2024). Classifier and classifier-free guidance are the canonical instance of steering a score by a conditional contrast (Dhariwal and Nichol 2021; Ho and Salimans 2022). Our same-state score residual is a same-state, image-conditioned instance of this same idea – a controlled contrast between two evaluations of the frozen model – but we apply it by directly optimizing the latent rather than by reweighting the sampling update.

Rectified flow and inversion. Diffusion and score-based models learn a noise-conditional vector field (Ho, Jain, and Abbeel 2020; Song et al. 2021); flow matching and rectified flow instead parameterize a direct, near-straight-line velocity field (Lipman et al. 2023; Liu, Gong, and Liu 2023), adopted by modern large-scale text-to-image transformers (Esser et al. 2024). Reconciling editing with this parameterization has motivated rectified-flow-specific inversion (Rout et al. 2025a) and inversion-free trajectory construction between two text conditions (Kulikov et al. 2025). We never invert a real image and never reconstruct a source trajectory; we optimize the model's own currently-generated latent using a same-state contrast, exploiting the straight-line structure of the ODE to keep that contrast a clean directional signal (§ Method).

Style-similarity metrics. Style similarity is commonly measured with a contrastive style descriptor (CSD) trained to separate artists (Somepalli et al. 2024). A recent calibration study shows that raw CSD cosine similarity gives negative discrimination gaps for a majority of a broad artist panel, and proposes a corrected CSLS-style readout (Frochte 2026). This finding motivates our calibration analysis (§ Results): because CSD conflates style with the reference's subject identity, it can penalize a stylistically excellent transfer whose prompt subject differs from the reference's subject. *Scope.* We do not run externally published style-transfer systems as competitor baselines in this paper; every comparison here is an internal ablation, a 40-style breadth benchmark, and this calibration analysis (see Limitations).

Method

Same-State Score Residual

Let $z_t \in \mathbb{R}^{L\times C}$ be the current noisy image tokens at sampler timestep t with noise level σ_t, and let p be a content prompt. The frozen model exposes a native image-conditioning path: it predicts velocity $\mathsf{v}_s(z_t, t, p, x_s)$ when shown a style reference image x_s as an image condition, and $\mathsf{v}_0(z_t, t, p)$ when that condition is removed. The two evaluations must share the same z_t, t, prompt metadata, and image-token IDs – otherwise their difference contains an uncontrolled state change and cannot be interpreted as guidance. We call this the *same-state contract*.

For the flow scheduler

$$z_\sigma = (1-\sigma)x_0 + \sigma\epsilon, \qquad \mathsf{v} = \epsilon - x_0, \tag{1}$$

we have $\hat{x}_0 = z_\sigma - \sigma\mathsf{v}$ and $\mathsf{s} = -(z_\sigma + (1-\sigma)\mathsf{v})/\sigma$. Shared state terms cancel between the two branches, giving the **score residual**

$$g_t \equiv \mathsf{s}_s - \mathsf{s}_0 = -\frac{1-\sigma}{\sigma}\left(\mathsf{v}_s - \mathsf{v}_0\right). \tag{2}$$

The implementation resolves σ from the scheduler table and sets the correction to zero when the inverse conversion is undefined at the degenerate endpoint. If the two branches estimate conditional densities p_s and p_0, Eq. 2 is an image analogue of classifier-free guidance:

$$\begin{aligned} g_t &= \nabla_{z_t}\log p_s(z_t \mid p, x_s) - \nabla_{z_t}\log p_0(z_t \mid p) \\ &= \nabla_{z_t}\log\frac{p_s}{p_0}. \end{aligned} \tag{3}$$

The frozen model – not a replaced attention feature, not a learned adapter – therefore defines what the reference asks the current sample to change.

Latent Optimization

We optimize *only* the current latent z_t; no network weights, no adapters, and no reference tensors are injected into attention. The without-reference branch predicts $\hat{x}_0^0 = z_t - \sigma\mathsf{v}_0(z_t)$ with gradients retained through v_0; the with-reference branch is detached. The goal-matching loss is

$$\mathcal{L}_{\text{goal}} = \left\| \mathcal{N}\left[-\frac{1-\sigma}{\sigma}\big(\mathsf{v}_0(z_t) - \text{sg}[\mathsf{v}_s(z_t)]\big)\right]\right\|_2^2, \tag{4}$$

where $\mathcal{N}$ removes the timestep scalar magnitude while preserving field direction and relative token energy, and $\text{sg}[\cdot]$ denotes a stop-gradient. Two light regularizers keep the update inside the model's valid local neighborhood: a distance term $\mathcal{L}_{\text{dist}}$ that discourages the optimized latent from drifting far from its pre-update value, and a content term $\mathcal{L}_{\text{content}}$ that anchors the prompt-conditioned prediction. The full per-step objective is

$$\mathcal{L} = \mathcal{L}_{\text{goal}} + \lambda_d\mathcal{L}_{\text{dist}} + \lambda_p\mathcal{L}_{\text{content}}. \tag{5}$$

AdamW updates only z_t for a handful of inner iterations at every outer denoising step (five inner iterations, 25 outer steps, start time 1000 – a high-noise start so the residual is exposed before content has crystallized); with zero weight decay it is numerically identical to Adam, and weight decay is exposed only as an explicit optimizer ablation (§ Results). No warm-start color-matching or style-transport heuristic is used; the method described here *is* the full method.

One step, in full. (1) Save the post-scheduler latent z_t and resolve its exact σ. (2) Evaluate v_s and v_0 at that same state; form g_t via Eq. 2. (3) Update z_t for a few AdamW iterations by minimizing Eq. 5. (4) Advance the original flow sampler with the optimized z_t.

Why rectified flow matters. The straight-line ODE of rectified flow makes the same-state residual a clean directional signal end to end: because the velocity field is close to constant along the sampling trajectory, a same-state contrast measured at one point remains informative for the local update rather than describing a curvature effect specific to that timestep. We exploit the model's own image conditioning rather than replacing internal attention features, and we optimize online, per sample, rather than training a generator or an adapter.

Experimental Setup

Model. All experiments use FLUX.2 [klein] 9B, a frozen rectified-flow text-to-image transformer with a native image-conditioning path; we never fine-tune, LoRA-adapt, or otherwise update its weights.

Benchmark. Our benchmark pairs 40 style reference images with content prompts drawn from five prompt classes (panda, castle, cat, sunflowers, fisherman). Every style is evaluated against at least one content prompt with at least two random seeds (drawn from $\{64, 65, 66\}$); the resulting 91 matched (style, prompt, seed) cells span all 40 styles.

Metrics. We report CSD (Somepalli et al. 2024) and a CSLS-calibrated CSD+ readout (Frochte 2026) for style similarity, CLIP-T (Radford et al. 2021) for prompt adherence, DINO (Oquab et al. 2024) style similarity, VGG-Gram distance for texture-statistics similarity, and a DINO self-similarity correlation with the reference as a layout-leakage proxy (lower is better – it measures how much of the reference's own composition or structure has copied into the output).

No external competitor baselines. We do not compare against externally published style-transfer systems (see Limitations). Every comparison in this paper is an internal ablation (§ Results), a 40-style breadth benchmark (§ Results), and a calibration analysis (§ Results).

Results

Breadth (Headline)

Table 1 reports per-style aggregate metrics across all 40 styles (91 matched cells, multi-seed). Per-style mean CSD is 0.600 (95% bootstrap CI $[0.562, 0.638]$), CLIP-T is 0.311 ($[0.304, 0.318]$, solid prompt adherence), layout-leak is 0.356 ($[0.331, 0.381]$), and VGG-Gram distance is 3.47. At the level of individual cells rather than per-style aggregates ($n = 91$), CSD has mean 0.597 and standard deviation 0.130 – consistent with, but distinct from, the per-style-aggregated headline number above. Qualitatively (Fig. 1) transfers are uniformly sharp and content-preserving across impasto oil, pop-art halftone, B&W cross-hatch engraving, woodcut, anime/digital painting, and expressionist brushwork: the prompt's subject stays legible while the reference's style is faithfully applied.

Table 1: Breadth benchmark: per-style mean over all 40 styles / 91 matched cells, multi-seed $\{64, 65, 66\}$. CIs are 95% bootstrap intervals over per-style means; Gram is a point estimate (no bootstrap CI computed).

Metric	Mean	95% CI
CSD↑ (style similarity)	0.600	[0.562, 0.638]
CLIP-T↑ (prompt adherence)	0.311	[0.304, 0.318]
Layout-leak↓	0.356	[0.331, 0.381]
VGG-Gram↓	3.47	–

Style Metrics Are Uncalibrated – Read Qualitatively

Raw CSD systematically under-rates excellent transfers because it entangles style with *subject* similarity to the reference, not style alone. Figure 2 shows two clean examples. Style 25's reference is a black-and-white cross-hatch engraving of a hooded archer; applied to a sunflower-field prompt, the transfer is a flawless engraving of a sunflower field at sunset – yet it scores CSD only 0.49, because the archer's subject (a person, a bow) has nothing in common with the generated sunflower field. Style 10's reference is an expressionist thick-impasto portrait of a bearded man; applied to a castle-by-the-sea prompt, the transfer is an excellent bold-brushstroke castle painting – yet it scores CSD only 0.381, again because the reference's subject (a person) and the generated subject (a building) differ. This matches the calibration critique of Frochte (2026), who show raw CSD gives negative discrimination gaps for 23 of 91 tested artists. We therefore treat mean CSD 0.600 as a *lower bound* on true transfer quality, report the CSLS-calibrated CSD+ readout (Frochte 2026) alongside raw CSD as a complementary diagnostic, and rely on qualitative evaluation (Figs. 1, 2) for the headline quality claim.

The Minimal Form Is Sufficient (Negative Ablations)

We ablate three "obvious" additions to the plain score residual. All ablation numbers in Table 2 come from a smaller matched matrix used specifically for controlled, paired comparisons (not the 40-style breadth benchmark of Table 1), and all deltas should be read against the ± 0.11 CSD seed-noise floor measured on textured styles (§ Limitations).

Content-affinity carrier. A natural next step multiplicatively reweights the score residual toward tokens the current content latent already attends to – a transport carrier in the spirit of appearance-transport methods. Across five independent probes it is inert or, at best, style-conditional: a matched 3×3 ablation matrix gives carrier CSD 0.5935 against 0.5993 for the plain residual (a wash, slightly worse on Gram/layout-leak too); a flat carrier-strength sweep $\{0.5, 0.75, 1.0\}$ never beats the plain residual ($0.5912/0.5935/0.5994$ against

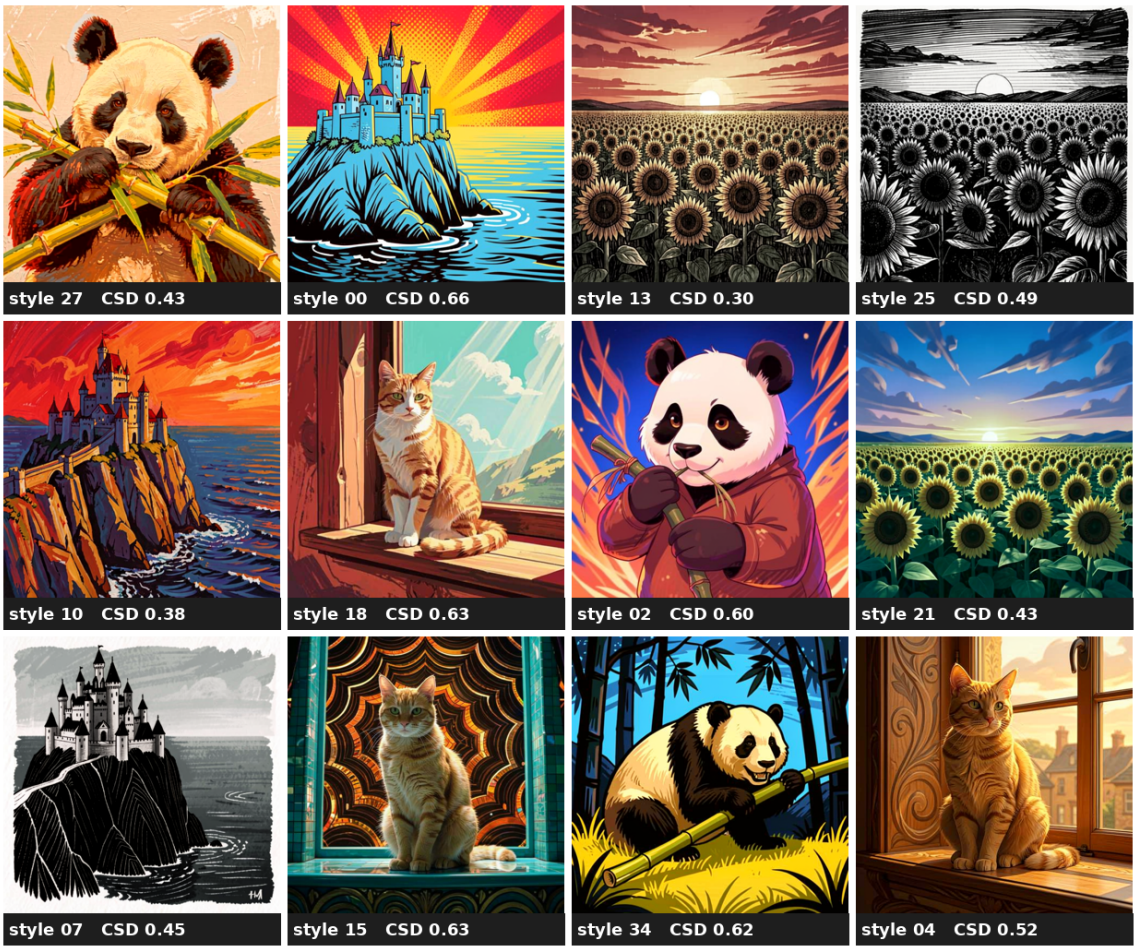


Figure 1: Twelve of the 40 benchmark styles (of 91 total matched cells), each labeled with its per-image CSD score. The method transfers sharp, content-preserving style across radically different mark-making – impasto oil, pop-art halftone, B&W cross-hatch engraving, woodcut, anime/digital painting, and expressionist brushwork – while the labeled CSD scores illustrate the calibration problem discussed in § "Style Metrics Are Uncalibrated": several of the visually strongest transfers (*e.g.* style 10, CSD 0.38; style 13, CSD 0.30) score no higher, or lower, than weaker ones.

0.5993); a timestep-scheduled carrier strength (const, back-loaded, front-loaded, $\sqrt{\cdot}$, cosine) changes CSD by less than 0.02 in every case, inside the seed-noise floor; the calibrated CSD+ readout does not rescue a carrier win, with two justified evaluation configurations disagreeing even in *direction* (-0.011 / -0.029 on the same style); and a method-delta LPIPS diagnostic shows the carrier changes the image substantially on heavily textured styles (mean 0.158, up to 0.560 on the most textured style, against 0.099 on a flatter style) without any accompanying quality gain on any of the four measured axes. The honest characterization is that the carrier is mostly inert, and on textured styles it perturbs the image without a robust quality gain – not that it is uniformly a no-op. Mechanistically, a single multiplicative term conflates structure-preservation with appearance-transport, which likely explains why it helps on no axis consistently.

Timestep scheduling. Scheduling the injection strength across the denoising trajectory (protecting content early and pushing style late, or the reverse) produces no effect distinguishable from seed noise (Table 2). Timing and magnitude of the residual are not a lever in this setting.

Latent optimizer. Switching between Adam and AdamW with weight decay $\{0, 10^{-4}, 10^{-3}\}$ moves CSD by only 0.01–0.05, consistent with AdamW at zero decay being numerically Adam. This is a robustness check, not a lever: the method is insensitive to this choice.

Limitations

No external published baseline. This is a scope choice, not an oversight: we compensate with the negative ablations (§ Results), 40-style breadth (§ Results), and the calibration analysis (§ Results). Reviewers may reasonably want a published external comparison; we regard this as addable future work rather than as evidence against the present claims, which are about the sufficiency of the minimal mechanism and the behavior of its natural extensions, not about beating

Table 2: Matched-subset ablations (paired probes; not the 40-style benchmark). Every axis is null: no configuration improves over the plain score residual beyond the ± 0.11 seed-noise floor.

Ablation axis	Configurations tested	CSD (matched subset)
Score residual (base)	matched 3×3 matrix	0.5993
+ Content-affinity carrier	matched 3×3 matrix	0.5935
Carrier strength	$\{0.5,\ 0.75,\ 1.0\}$	0.5912 / 0.5935 / 0.5994
Timestep schedule	const / back / front / sqrt / cosine	$\lvert\Delta\rvert < 0.02$ vs. const
Latent optimizer	Adam, AdamW wd $\{0, 10^{-4}, 10^{-3}\}$	Δ 0.01–0.05



Style 25 reference (archer) | Sunflowers, CSD 0.49

Style 10 reference (portrait) | Castle, CSD 0.381

Figure 2: CSD under-rating. Top: a B&W cross-hatch engraving reference whose subject (an archer) differs from the sunflower-field prompt still transfers the engraving style flawlessly, yet scores CSD only 0.49. Bottom: an expressionist thick-impasto portrait reference transfers convincingly onto a castle-by-the-sea prompt, yet scores CSD only 0.381. Both transfers are visually excellent; the low scores reflect subject mismatch with the reference, not a style-transfer failure.

a specific competitor.

ArtFID is not soundly computable at this scale. Standard ArtFID requires a style-side Fréchet distance whose covariance is rank-deficient when the number of style references (40) is far smaller than the Inception feature dimension; we therefore report per-image style/content metrics and CSD+ instead of an unstable ArtFID number.

Content-LPIPS needs a no-style baseline set. A content-preservation LPIPS metric requires a matched no-style generation for every prompt, which we have not yet generated; this is queued future work.

Seed variance is large on textured styles. CSD varies by up to ± 0.11 across seeds on heavily textured styles (*e.g.* style 27); ablation deltas smaller than this floor are noise, not signal, and we interpret Table 2 accordingly throughout § Results.

Conclusion

On a frozen rectified-flow transformer, training-free style transfer needs only the model's native same-state image-CFG score residual, applied by optimizing the current latent. Added transport machinery, timestep scheduling, and optimizer choice do not help beyond seed noise; the minimal form is sufficient and, read qualitatively – since raw CSD systematically under-rates it – strong and content-preserving across 40 styles. We view the negative ablations as being as important to report as the positive result: they mark exactly which extensions are not worth building on this mechanism, and where a future differentiator would need to look instead.

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